EXPERIMENT 13

13.1.1//Fibonacci sequence using module

ALGORITHM:

1. **Start**
2. **Input**
   - Read an integer `n` from the user.
3. **Check if input is valid**
   - If `n ≤ 0`, print:
     `"Please enter a positive integer"`
   - Else, proceed to generate the Fibonacci sequence.
4. **Call function** `generate_fibonacci_sequence(n)`

Function :

5. **Check for negative input**
   - If `max_value < 0`, return an empty list.
6. **Initialize variables**
   - Create an empty list `seq`
   - Set `a = 0`
   - Set `b = 1`
7. **Generate Fibonacci sequence**
   - While `a ≤ max_value`, do:
     - Add `a` to `seq`
     - Update values:
       - `a = b`
       - `b = a + b` (using tuple swap: `a, b = b, a + b`)
8. **Return result**
   - Return the list `seq`

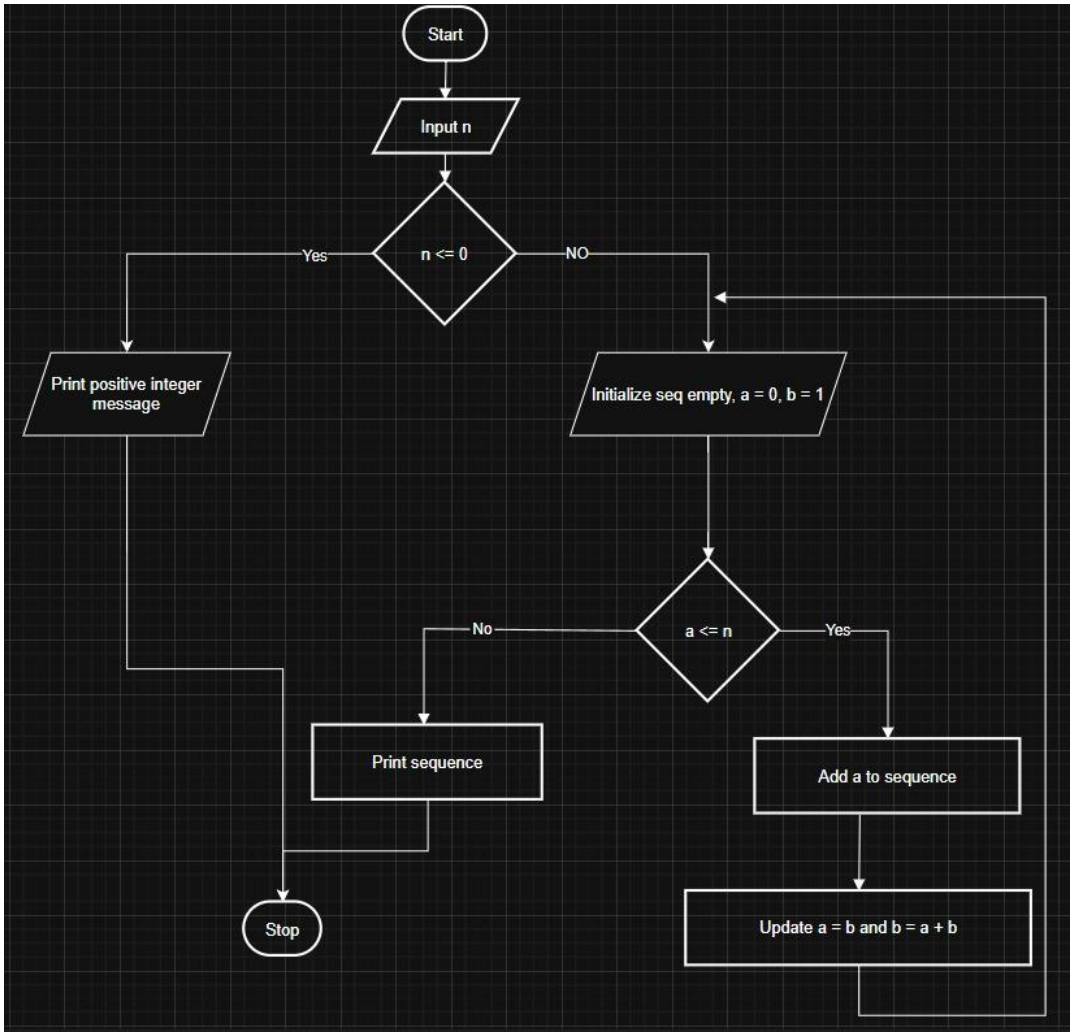
Back to Main Program

9. **Store result**
   - Save returned list in `fib_series`
10. **Output**

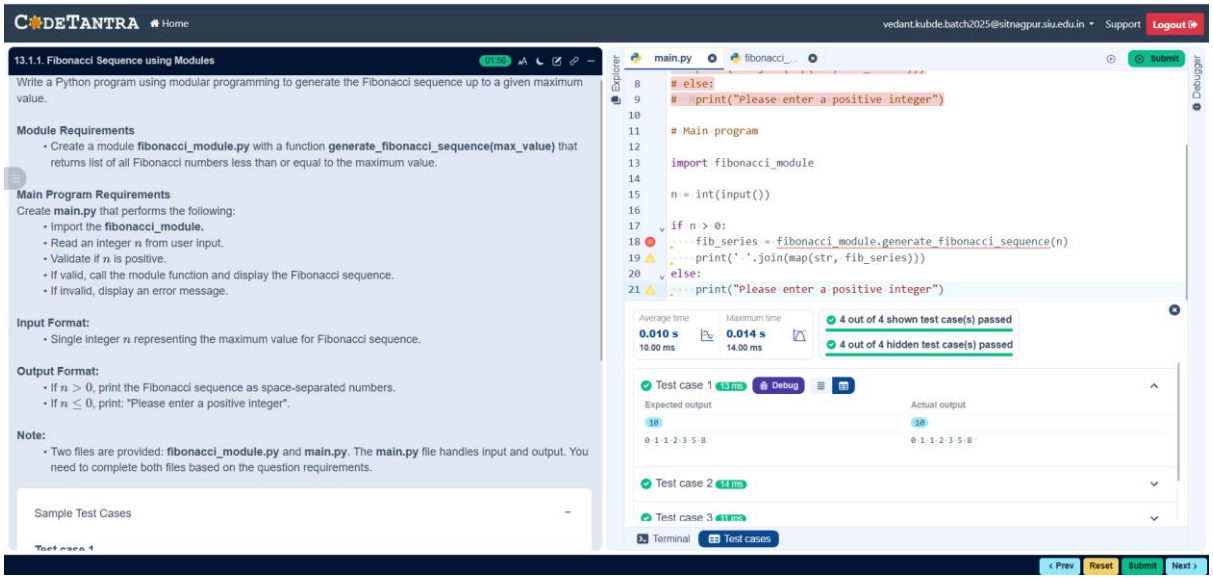
- Convert all elements to string
- Join them with spaces
- Print the sequence

11. **End**

FLOWCHART:


Start
Input n
n <= 0
Yes
NO
Print positive integer message
Initialize seq empty, a = 0, b = 1
a <= n
No
Yes
Print sequence
Add a to sequence
Stop
Update a = b and b = a + b

CODETANTRA:


CODETANTRA
Home
vedant.kubde.batch2025@sitnagpur.siu.edu.in
Support
Logout
13.1.1. Fibonacci Sequence using Modules
Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.
Module Requirements
• Create a module fibonacci_module.py with a function generate_fibonacci_sequence(max_value) that returns list of all Fibonacci numbers less than or equal to the maximum value.
Main Program Requirements
Create main.py that performs the following:
• Import the fibonacci_module.
• Read an integer n from user input.
• Validate if n is positive.
• If valid, call the module function and display the Fibonacci sequence.
• If invalid, display an error message.
Input Format:
• Single integer n representing the maximum value for Fibonacci sequence.
Output Format:
• If n > 0, print the Fibonacci sequence as space-separated numbers.
• If n ≤ 0, print: "Please enter a positive integer".
Note:
• Two files are provided: fibonacci_module.py and main.py. The main.py file handles input and output. You need to complete both files based on the question requirements.
Sample Test Cases
main.py
fibonacci_...
Submit
# else:
# print("Please enter a positive integer")
# Main program
import fibonacci_module
n = int(input())
if n > 0:
fib_series = fibonacci_module.generate_fibonacci_sequence(n)
print(' '.join(map(str, fib_series)))
else:
print("Please enter a positive integer")
Average time
0.010 s
10.00 ms
Maximum time
0.014 s
14.00 ms
4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
Debug
Expected output
Actual output
Test case 2
Test case 3
Terminal
Test cases
Prev
Reset
Submit
Next